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4

Rationality and Common Knowledge

In this chapter we study the implications of imposing the assumptions of rationality as well as common knowledge of rationality. We derive and explore some solution concepts that result from these two assumptions and seek to understand the restrictions that each of the two assumptions imposes on the way in which players will play games.

4.1 Dominance in Pure Strategies

It will be useful to begin by introducing some new notation. We denoted the payoff of a player i from a profile of strategies $s = (s_1, s_2, \ldots, s_{i-1}, s_i, s_{i+1}, \ldots, s_n)$ as $v_i(s)$. It will soon be very useful to refer specifically to the strategies of a player's opponents in a game. For example, the actions chosen by the players who are *not* player i are denoted by the profile

$$(s_1, s_2, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n) \in S_1 \times S_2 \times \cdots \times S_{i-1} \times S_{i+1} \times \cdots \times S_n.$$

To simplify we will hereafter use a common shorthand notation as follows: We define $S_{-i} \equiv S_1 \times S_2 \times \cdots \times S_{i-1} \times S_{i+1} \times \cdots \times S_n$ as the set of all the strategy sets of *all players who are not* player i. We then define $s_{-i} \in S_{-i}$ as a particular possible profile of strategies for all players who are not i. Hence we can rewrite the payoff of player i from strategy s as $v_i(s_i, s_{-i})$, where $s = (s_i, s_{-i})$.

4.1.1 Dominated Strategies

The Prisoner's Dilemma was easy to analyze: each of the two players has an action that is best *regardless* of what his opponent chooses. Suggesting that each player will choose this action seems natural because it is consistent with the basic concept of rationality. If we assume that the players are rational, then we should expect them to choose whatever they deem to be best for them. If it turns out that a player's best strategy does not depend on the strategies of his opponents then we should be all the more confident that he will choose it.

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It is not too often that we will find ourselves in situations in which we have a best action that does not depend on the actions of our opponents. We begin, therefore, with a less demanding concept that follows from rationality. In particular consider the strategy mum in the Prisoner's Dilemma:

		Player 2	
		M	F
Player 1	M	$-2, -2$	$-5, -1$
	F	$-1, -5$	$-4, -4$

As we argued earlier, playing M is worse than playing F for each player *regardless* of what the player's opponent does. What makes it unappealing is that there is another strategy, F, that is better than M regardless of what one's opponent chooses. We say that such a strategy is *dominated*. Formally we have

Definition 4.1 Let $s_i \in S_i$ and $s_i' \in S_i$ be possible strategies for player i. We say that s_i' is **strictly dominated** by s_i if for any possible combination of the other players' strategies, $s_{-i} \in S_{-i}$, player i's payoff from s_i' is strictly less than that from s_i. That is,

$$v_i(s_i, s_{-i}) > v_i(s_i', s_{-i}) \quad \text{for all } s_{-i} \in S_{-i}.$$

We will write $s_i \succ_i s_i'$ to denote that s_i' is strictly dominated by s_i.

Now that we have a precise definition for a dominated strategy, it is straightforward to draw an obvious conclusion:

Claim 4.1 *A rational player will never play a strictly dominated strategy.*

This claim is obvious. If a player plays a dominated strategy then he cannot be playing optimally because, by the definition of a dominated strategy, the player has another strategy that will yield him a higher payoff *regardless of the strategies of his opponents.* Hence knowledge of the game implies that a player should recognize dominated strategies, and rationality implies that these strategies will be avoided.

When we apply the notion of a dominated strategy to the Prisoner's Dilemma we argue that each of the two players has one dominated strategy that he should never use, and hence each player is left with one strategy that is not dominated. Therefore, for the Prisoner's Dilemma, rationality alone is enough to offer a prediction about which outcome will prevail: (F, F) is this outcome.

Many games, however, will not be as special as the Prisoner's Dilemma, and rationality alone will not suggest a clear-cut, unique prediction. As an example, consider the following advertising game. Two competing brands can choose one of three marketing campaigns—low (L), medium (M), and high (H)—with payoffs given by the following matrix:

		Player 2		
		L	M	H
	L	6, 6	2, 8	0, 4
Player 1	M	8, 2	4, 4	1, 3
	H	4, 0	3, 1	2, 2

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It is easy to see that each player has one dominated strategy, which is L. However, neither M nor H is dominated. For example, if player 2 plays M then player 1 should also play M, while if player 2 plays H then player 1 should also play H. Hence rationality alone does not offer a unique prediction. It is nonetheless worth spending some time on the extreme cases in which it does.

4.1.2 Dominant Strategy Equilibrium

Because a strictly dominated strategy is one to avoid at all costs,[1] there is a counterpart strategy, represented by F in the Prisoner's Dilemma, that would be desirable. This is a strategy that is *always* the best thing you can do, regardless of what your opponents choose. Formally we have

Definition 4.2 $s_i \in S_i$ is a **strictly dominant strategy** for i if every other strategy of i is strictly dominated by it, that is,

$$v_i(s_i, s_{-i}) > v_i(s_i', s_{-i}) \quad \text{for all } s_i' \in S_i, \quad s_i' \neq s_i, \quad \text{and all } s_{-i} \in S_{-i}.$$

If, as in the Prisoner's Dilemma, every player had such a wonderful dominant strategy, then it would be a very sensible predictor of behavior because it follows from rationality alone. We can introduce this simple idea as our first solution concept:

Definition 4.3 The strategy profile $s^D \in S$ is a **strict dominant strategy equilibrium** if $s_i^D \in S_i$ is a strict dominant strategy for all $i \in N$.

This gives a formal definition for the outcome "both players fink," or (F, F), in the Prisoner's Dilemma: it is a dominant strategy equilibrium. In this equilibrium the payoffs are $(-4, -4)$ for players 1 and 2, respectively.

Caveat Be careful not to make a common error by referring to the pair of payoffs $(-4, -4)$ as the solution. The solution should always be described as the strategies that the players will choose. Strategies are a set of actions by the players, and payoffs are a result of the outcome. When we talk about predictions, or equilibria, we will always refer to *what players do* as the equilibrium, not their payoffs.

Using this solution concept for any game is not that difficult. It basically requires that we identify a strict dominant strategy for each player and then use this profile of strategies to predict or prescribe behavior. If, as in the Prisoner's Dilemma, we are lucky enough to find a dominant strategy equilibrium for other games, then this solution concept has a very appealing property:

Proposition 4.1 *If the game* $\Gamma = \langle N, \{S_i\}_{i=1}^n, \{v_i\}_{i=1}^n \rangle$ *has a strictly dominant strategy equilibrium* s^D*, then* s^D *is the unique dominant strategy equilibrium.*

This proposition is rather easy to prove, and that proof is left as exercise 4.1 at the end of the chapter.

1. This is a good point at which to stop and reflect on a very simple yet powerful lesson. In any situation, look first for your dominated strategies and avoid them!

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4.1.3 Evaluating Dominant Strategy Equilibrium

Proposition 4.1 is very useful in addressing one of our proposed criteria for evaluating a solution concept: when it exists, the strict-dominance solution concept guarantees uniqueness. However, what do we know about existence? A quick observation will easily convince you that this is a problem.

Consider the Battle of the Sexes game introduced in Section 3.3 and described again in the following matrix:

		Chris	
		O	F
Alex	O	2, 1	0, 0
	F	0, 0	1, 2

Neither player has a dominated strategy, implying that neither has a dominant strategy either. The best strategy for Chris depends on what Alex chooses and vice versa. Thus if we stick to the solution concept of strict dominance we will encounter games, in fact many of them, for which there will be no equilibrium. This unfortunate conclusion implies that the strict-dominance solution concept will often fail to predict the choices that players ought to, or will, choose in many games.

Regarding the invariance criterion, the strict-dominance solution concept does comply. From definition 4.2, s^D is a strictly dominant strategy equilibrium if and only if $v_i(s_i^D, s_{-i}) > v_i(s_i', s_{-i})$ for all $s_i' \in S_i$ and all $s_{-i} \in S_{-i}$. Because the inequality is strict, we can find a small enough value $\varepsilon > 0$ such that if we either add or subtract ε from any payoff $v_i(s_i, s_{-i})$ the inequality will still hold.

We now turn to the Pareto criterion of equilibrium outcomes when a strict-dominance solution exists. The Prisoner's Dilemma has an equilibrium prediction using the strict-dominance solution concept, so we can evaluate the efficiency properties of the unique strictly dominant strategy equilibrium for that game. It is easy to see that the outcome prescribed by this solution is *not Pareto optimal:* both players would be better off if they could each commit to play M, yet left to their own devices they will not do this. Of course, in other games the solution may be Pareto optimal (see, for example, the "altruistic" Prisoner's Dilemma in Section 3.3).

The failure of Pareto optimality is *not* a failure of the solution concept. The assumption that players are rational causes them to fink in the Prisoner's Dilemma if we restrict attention to self-enforcing behavior. The failure of Pareto optimality implies that the players would benefit from modifying the environment in which they find themselves to create other enforcement mechanisms—for example, creating a "mafia" with norms of conduct that enforce implicit agreements so as to punish those who fink.

To explicitly see how this can work, imagine that a mafia member who finks on another member is very seriously reprimanded, which will change the payoff structure of the Prisoner's Dilemma if he is caught. If the pain from mafia punishment is equivalent to z, then we have to subtract z units of payoff for each player who finks. The "mafia-modified" Prisoner's Dilemma is represented by the following matrix:

		Player 2	
		M	F
Player 1	M	$-2, -2$	$-5, -1-z$
	F	$-1-z, -5$	$-4-z, -4-z$

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If z is strictly greater than 1 then this punishment will be enough to flip our predicted equilibrium outcome of the game because then M becomes the strict dominant strategy (and (M, M) is Pareto optimal).

This example demonstrates that "institutional design," which changes the game that players play, can be very useful in affecting the well-being of players. By introducing this external enforcer, or institution, we are able to get the players to choose outcomes that make them both better off compared to what they can achieve without the additional institution. Moreover, notice that if the players believe that the mafia will enforce the code of conduct then there is no need to actually enforce it—the players choose not to fink, and the enforcement of punishments need not happen. However, we need to be suspicious of whether such an institution will be *self-enforcing,* that is, whether the mafia will indeed enforce the punishments. And, for it to be self-enforcing, we need to model the behavior of potential punishers and whether they themselves will have the selfish incentives to carry out the enforcement activities. This is something we will explore at length when we consider multistage and repeated games in Chapters Chapters 9 and 10.

Remark A related notion is that of **weak dominance.** We say that s_i' is *weakly dominated* by s_i if, for any possible combination of the other players' strategies, player i's payoff from s_i' is weakly less than that from s_i. That is,

$$v_i(s_i, s_{-i}) \geq v_i(s_i', s_{-i}) \quad \text{for all } s_{-i} \in S_{-i}.$$

This means that for some $s_{-i} \in S_{-i}$ this weak inequality may hold strictly, while for other $s'_{-i} \in S_{-i}$ it will hold with equality. We define a strategy to be **weakly dominant** in a similar way. This is still useful because if we can find a dominant strategy for a player, be it weak or strict, this seems like the most obvious thing to prescribe. An important difference between weak and strict dominance is that if a weakly dominant equilibrium exists, it need not be unique. To show this is left as exercise 4.2.

4.2 Iterated Elimination of Strictly Dominated Pure Strategies

As we saw in the previous chapter, our requirement that players be rational implied two important conclusions:

1. A rational player will never play a dominated strategy.
2. If a rational player has a strictly dominant strategy then he will play it.

We used this second conclusion to define the solution concept of strict dominance, which is very appealing because, when it exists, it requires only rationality as its driving force. A drawback of the dominant strategy solution concept is, however, that it will often fail to exist. Hence if we wish to develop a predictive theory of behavior in games then we must consider alternative approaches that will apply to a wide variety of games.

4.2.1 Iterated Elimination and Common Knowledge of Rationality

We begin with the premise that players are rational, and we build on the first conclusion in the previous section, which claims that a rational player will never play a

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dominated strategy. This conclusion is by itself useful in that it rules out what players *will not do*. As a result, we conclude that rationality tells us which strategies will never be played.

Now turn to another important assumption introduced earlier: the structure of the game and the rationality of the players are common knowledge among the players. The introduction of *common knowledge of rationality* allows us to do much more than identify strategies that rational players will avoid. If indeed all the players *know* that each player will never play a strictly dominated strategy, they can effectively ignore those strictly dominated strategies that their opponents will never play, and their opponents can do the same thing. If the original game has some players with some strictly dominated strategies, then all the players know that *effectively* they are facing a "smaller" restricted game with fewer total strategies.

This logic can be taken further. Because it is *common knowledge* that all players are rational, then everyone knows that everyone knows that the game is effectively a smaller game. In this smaller restricted game, everyone knows that players will not play strictly dominated strategies. In fact we may indeed find additional strategies that are dominated in the restricted game that *were not dominated* in the original game. Because it is common knowledge that players will perform this kind of reasoning again, the process can continue until no more strategies can be eliminated in this way.

To see this idea more concretely, consider the following two-player finite game:

		Player 2		
		L	*C*	*R*
	U	4, 3	5, 1	6, 2
Player 1	*M*	2, 1	8, 4	3, 6
	D	3, 0	9, 6	2, 8

(4.1)

A quick observation reveals that there is no strictly dominant strategy, neither for player 1 nor for player 2. Also note that there is no strictly dominated strategy for player 1. There is, however, a strictly dominated strategy for player 2: the strategy C is strictly dominated by R because $2 > 1$ (row U), $6 > 4$ (row M), and $8 > 6$ (row D). Thus, because this is common knowledge, both players know that we can effectively eliminate the strategy C from player 2's strategy set, which results in the following reduced game:

		Player 2	
		L	*R*
	U	4, 3	6, 2
Player 1	*M*	2, 1	3, 6
	D	3, 0	2, 8

In this reduced game, both M and D are strictly dominated by U for player 1, allowing us to perform a second round of eliminating strategies, this time for player 1. Eliminating these two strategies yields the following trivial game:

		Player 2	
		L	*R*
Player 1	*U*	**4, 3**	6, 2

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in which player 2 has a strictly dominated strategy, playing R. Thus for this example the iterated process of eliminating dominated strategies yields a unique prediction: the strategy profile we expect these players to play is (U, L), giving the players the payoffs of (4, 3).

As the example demonstrates, this process of **iterated elimination of strictly dominated strategies** (IESDS) builds on the assumption of common knowledge of rationality. The first step of iterated elimination is a consequence of player 2's rationality; the second stage follows because players *know* that players are rational; the third stage follows because players *know that players know* that they are rational, and this ends in a unique prediction.

More generally we can apply this process to games in the following way. Let S_i^k denote the strategy set of player i that survives k rounds of IESDS. We begin the process by defining $S_i^0 = S_i$ for each i, the original strategy set of player i in the game.

Step 1: Define $S_i^0 = S_i$ for each i, the original strategy set of player i in the game, and set $k = 0$.

Step 2: Are there players for whom there are strategies $s_i \in S_i^k$ that are strictly dominated? If yes, go to step 3. If not, go to step 4.

Step 3: For all the players $i \in N$, remove any strategies $s_i \in S_i^k$ that are strictly dominated. Set $k = k + 1$, and define a new game with strategy sets S_i^k that *do not include* the strictly dominated strategies that have been removed. Go back to step 2.

Step 4: The remaining strategies in S_i^k are reasonable predictions for behavior.

In this chapter we refrain from giving a precise mathematical definition of the process because this requires us to consider richer behavior by the players, in particular, allowing them to choose randomly between their different pure strategies. We will revisit this approach briefly when such stochastic play, or **mixed strategies,** is introduced later.[2]

Using the process of IESDS we can define a new solution concept:

Definition 4.4 We will call any strategy profile $s^{ES} = (s_1^{ES}, \ldots, s_n^{ES})$ that survives the process of IESDS an **iterated-elimination equilibrium.**

Like the concept of a strictly dominant strategy equilibrium, the iterated-elimination equilibrium starts with the premise of rationality. However, in addition to rationality, IESDS requires a lot more: common knowledge of rationality. We will discuss the implications of this requirement later in this chapter.

4.2.2 Example: Cournot Duopoly

Recall the Cournot duopoly example we introduced in Section 3.1.2, but consider instead a simpler example of this problem in which the firms have linear rather than quadratic costs: the cost for each firm for producing quantity q_i is given by $c_i(q_i) = 10q_i$ for $i \in \{1, 2\}$. (Using economics jargon, this is a case of constant marginal cost

2. Just to satisfy your curiosity, think of the Battle of the Sexes, and imagine that Chris can pull out a coin and flip between the decision of opera or football. This by itself introduces a "new" strategy, and we will exploit such strategies to develop a formal definition of IESDS.

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equal to 10 and no fixed costs.) Let the demand be given by $p(q) = 100 - q$, where $q = q_1 + q_2$.

Consider first the profit (payoff) function of firm 1:

$$\begin{aligned} v_1(q_1, q_2) &= \overbrace{(100 - q_1 - q_2)q_1}^{\text{Revenue}} - \overbrace{10q_1}^{\text{Costs}} \\ &= 90q_1 - q_1^2 - q_1q_2. \end{aligned}$$

What should firm 1 do? If it knew what quantity firm 2 will choose to produce, say some value of q_2, then the profits of firm 1 would be maximized when the first-order condition is satisfied, that is, when $90 - 2q_1 - q_2 = 0$. Thus, for any given value of q_2, firm 1 maximizes its profits when it sets its own quantity according to the function

$$q_1 = \frac{90 - q_2}{2}. \tag{4.2}$$

Though it is true that the choice of firm 1 depends on what it believes firm 2 is choosing, equation (4.2) implies that firm 1 will *never* choose to produce more than $q_1 = 45$. This follows from the simple observation that q_2 is never negative, in which case equation (4.2) implies that $q_1 \leq 45$. In fact, this is equivalent to showing that any quantity $q_1 > 45$ is strictly dominated by $q_1 = 45$. To see this, for any q_2, the profits from setting $q_1 = 45$ are given by

$$v_1(45, q_2) = (100 - 45 - q_2)45 - 450 = 2025 - 45q_2.$$

The profits from choosing any other q_1 are given by

$$v_1(q_1, q_2) = (100 - q_1 - q_2)q_1 - 10q_1 = 90q_1 - q_1q_2 - q_1^2.$$

Thus we can subtract $v_1(q_1, q_2)$ from $v_1(45, q_2)$ and obtain

$$\begin{aligned} v_1(45, q_2) - v_1(q_1, q_2) &= 2025 - 45q_2 - (90q_1 - q_1q_2 - q_1^2) \\ &= 2025 - q_1(90 - q_1) - q_2(45 - q_1). \end{aligned}$$

It is easy to check that for any $q_1 > 45$ this difference is positive regardless of the value of q_2.[3] Hence we conclude that any $q_1 > 45$ is strictly dominated by $q_1 = 45$.

It is easy to see that firm 2 faces exactly the same profit function, which implies that any $q_2 > 45$ is strictly dominated by $q_2 = 45$. This observation leads to our first round of iterated elimination: a rational firm produces no more than 45 units, implying that the effective strategy space that survives one round of elimination is $q_i \in [0, 45]$ for $i \in \{1, 2\}$.

We can now turn to the second round of elimination. Because $q_2 \leq 45$, equation (4.2) implies that firm 1 will choose a quantity no less than 22.5, and a symmetric argument applies to firm 2. Hence the second round of elimination implies that the surviving strategy sets are $q_i \in [22.5, 45]$ for $i \in \{1, 2\}$.

The next step of this process will reduce the strategy set to $q_i \in [22.5, 33\frac{3}{4}]$, and the process will continue on and on. Interestingly the set of strategies that survives

3. This follows because if $q_1 > 45$ then $q_1(90 - q_1) < 2025$ and $q_2(45 - q_1) \leq 0$ for any $q_2 \geq 0$.

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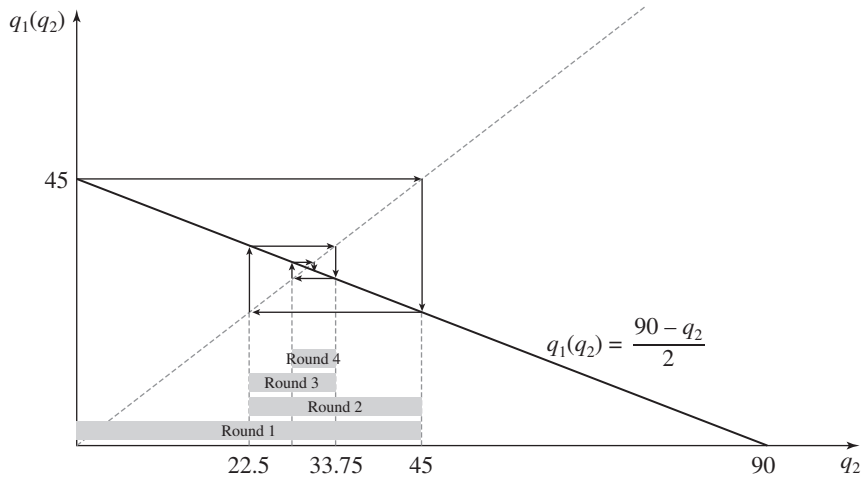

FIGURE 4.1 IESDS convergence in the Cournot game.

this process converges to a single quantity choice of $q_i = 30$. To see this, notice how we moved from one surviving interval to the next. We started by noting that $q_2 \geq 0$, and using equation (4.2) we found that $q_1 \leq 45$, creating the first-round interval of [0, 45]. Then, by symmetry, it follows that $q_2 \leq 45$, and using equation (4.2) again we conclude that $q_1 \geq 22.5$, creating the second-round interval [22.5, 45]. We can see this process graphically in Figure 4.1, where we use the upper (lower) end of the previous interval to determine the lower (upper) end of the next one. If this were to converge to an interval and not to a single point, then by the symmetry between both firms, the resulting interval for each firm would be $[q_{\min}, q_{\max}]$ that simultaneously satisfy two equations with two unknowns: $q_{\min} = \frac{90-q_{\max}}{2}$ and $q_{\max} = \frac{90-q_{\min}}{2}$. However, the only solution to these two equations is $q_{\min} = q_{\max} = 30$. Hence using IESDS for the Cournot game results in a unique predictor of behavior where $q_1 = q_2 = 30$, and each firm earns a profit of $v_1 = v_2 = 900$.

4.2.3 Evaluating IESDS

We turn to evaluate the IESDS solution concept using the criteria we introduced earlier. Start with existence and note that, unlike the concept of strict dominance, we can apply IESDS to *any* game by applying the algorithm just described. It does not require the existence of a strictly dominant strategy, nor does it require the existence of strictly dominated strategies. It is the latter characteristic, however, that gives this concept some bite: when strictly dominated strategies exist, the process of IESDS is able to say something about how common knowledge of rationality restricts behavior.

It is worth noting that this existence result is a consequence of assuming common knowledge of rationality. By doing so we are giving the players the ability to reason through the strategic implications of rationality, and to do so over and over again, while correctly anticipating that other players can perform the same kind of reasoning. Rationality alone does not provide this kind of reasoning.

It is indeed attractive that an IESDS solution always exists. This comes, however, at the cost of uniqueness. In the simple 3×3 matrix game described in (4.1) and the

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Cournot duopoly game, IESDS implied the survival of a unique strategy. Consider instead the Battle of the Sexes, given by the following matrix:

		Chris	
		O	F
Alex	O	2, 1	0, 0
	F	0, 0	1, 2

IESDS cannot restrict the set of strategies here for the simple reason that neither O nor F is a strictly dominated strategy for each player. As we can see, this solution concept can be applied (it exists) to any game, but it will often fail to provide a unique solution. For the Battle of the Sexes game, IESDS can only conclude that "anything can happen."

After analyzing the efficiency of the outcomes that can be derived from strict dominance in Section 4.1.3, you may have anticipated the possible efficiency of IESDS equilibria. An easy illustration can be provided by the Prisoner's Dilemma. IESDS leaves (F, F) as the unique survivor, or IESDS equilibrium, after only one round of elimination. As we already demonstrated, the outcome from (F, F) is not Pareto optimal. Similarly, both previous examples (the 3×3 matrix game in (4.1) and the Cournot game) provide further evidence that Pareto optimality need not be achieved by IESDS: In the 3×3 matrix example, both strategy profiles (M, C) and (D, C) yield higher payoffs for both players—(8, 4) and (9, 6), respectively—than the unique IESDS equilibrium, which yields (4, 3). For the Cournot game, producing $q_1 = q_2 = 30$ yields profits of 900 for each firm. If instead they would both produce $q_1 = q_2 = 20$ then each would earn profits of 1000. Thus common knowledge of rationality does not mean that players can guarantee the best outcome for themselves when their own incentives dictate their behavior.

On a final note, it is interesting to observe that there is a simple and quite obvious relationship between the IESDS solution concept and the strict-dominance solution concept:

Proposition 4.2 *If for a game* $\Gamma = \langle N, \{S_i\}_{i=1}^n, \{v_i\}_{i=1}^n \rangle$ s^* *is a* strict dominant strategy equilibrium, *then* s^* uniquely survives IESDS.

Proof If $s^* = (s_1^*, \ldots, s_n^*)$ is a strict dominant strategy equilibrium then, by definition, for every player i all other strategies s_i' are strictly dominated by s_i^*. This implies that after one stage of elimination we will be left with a single profile of strategies, which is exactly s^*, and this concludes the proof. ■

This simple proposition is both intuitive and straightforward. Because rationality is the only requirement needed in order to eliminate all strictly dominated strategies in one round, then if all strategies but one are strictly dominated for each and every player, both IESDS and strict dominance will result in the same outcome. This shows us that whenever strict dominance results in a unique outcome, then IESDS will result in the same unique outcome after one round. However, as we saw earlier, IESDS may offer a fine prediction when strict dominance does not apply. This is exactly what the extra assumption of common knowledge of rationality delivers: a more widely applicable solution concept. However, the assumption of common knowledge of rationality is far from innocuous. It requires the players to be, in some way, extremely

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intelligent and to possess unusual levels of foresight. For the most part, game theory relies on this strong assumption, and hence it must be applied to the real world with caution. Remember the rule about how assumptions drive conclusions: garbage in, garbage out.

4.3 Beliefs, Best Response, and Rationalizability

Both of the solution concepts we have seen so far, strict dominance and IESDS, are based on eliminating actions that players would never play. An alternative approach is to ask: what possible strategies might players choose to play and under what conditions? When we considered eliminating strategies that no rational player would choose to play, it was by finding some strategy that is *always* better or, as we said, that dominates the eliminated strategies. A strategy that cannot be eliminated, therefore, suggests that *under some conditions* this strategy is the one that the player may like to choose. When we qualify a strategy to be the best one a player can choose *under some conditions,* these conditions must be expressed in terms that are rigorous and are related to the game that is being played.

To set the stage, think about situations in which you were puzzled about the behavior of someone you knew. To consider his choice as irrational, or simply stupid, you would have to consider whether there is a way in which he could defend his action as a good choice. A natural way to determine whether this is the case is to simply ask him, "What were you thinking?" If the response lays out a plausible situation for which his choice was a good one, then you cannot question his rationality. (You may of course question his wisdom, or even his sanity, if his thoughts seem bizarre.)

This is a type of reasoning that we will formalize and discuss in this chapter. If a strategy s_i is not strictly dominated for player i then it must be that there are combinations of strategies of player i's opponents for which the strategy s_i is player i's best choice. This reasoning will allow us to justify or *rationalize* the choice of player i.

4.3.1 The Best Response

As we discussed early in Chapter 3, what makes a game different from a single-player decision problem is that once you understand the actions, outcomes, and preferences of a decision problem, then you can choose your best or optimal action. In a game, however, your optimal decision not only depends on the structure of the game, but it will often depend on what the other players are doing.

Take the Battle of the Sexes as an example:

		Chris	
		O	*F*
Alex	*O*	2, 1	0, 0
	F	0, 0	1, 2

As the matrix demonstrates, the best choice of Alex depends on what Chris will do. If Chris goes to the opera then Alex would rather go to the opera instead of going to the football game. If, however, Chris goes to the football game then Alex's optimal action is switched around.

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This simple example illustrates an important idea that will escort us throughout this book and that (one hopes) will escort you through your own decision making in strategic situations. In order for a player to be optimizing in a game, he has to choose a best strategy *as a response* to the strategies of his opponents. We therefore introduce the following formal definition:

Definition 4.5 The strategy $s_i \in S_i$ is player i's **best response** to his opponents' strategies $s_{-i} \in S_{-i}$ if

$$v_i(s_i, s_{-i}) \geq v_i(s_i', s_{-i}) \quad \forall s_i' \in S_i.$$

I can't emphasize enough how central this definition is to the concept of strategic behavior and rationality. In fact rationality implies that given any belief a player has about his opponents' behavior, he must choose an action that is best for him given his beliefs. That is,

Claim 4.2 *A rational player who believes that his opponents are playing some $s_{-i} \in S_{-i}$ will always choose a best response to s_{-i}.*

For instance, in the Battle of the Sexes, if Chris believes that Alex will go to the opera then Chris's best response is to go to the opera because

$$v_2(O, O) = 1 > 0 = v_2(O, F).$$

Similarly if Chris believes that Alex will go to the football game then Chris's best response is to go to the game as well.

There are some appealing relationships between the concept of playing a best response and the concept of dominated strategies. First, if a strategy s_i is strictly dominated, it means that some other strategy s_i' is *always* better. This leads us to the observation that the strategy s_i could not be a best response to *anything:*

Proposition 4.3 *If s_i is a* strictly dominated strategy *for player i, then it* cannot be a best response *to any $s_{-i} \in S_{-i}$.*

Proof If s_i is strictly dominated, then there exists some $s_i' \succ_i s_i$ such that $v_i(s_i', s_{-i}) > v_i(s_i, s_{-i})$ for all $s_{-i} \in S_{-i}$. But this in turn implies that there is no $s_{-i} \in S_{-i}$ for which $v_i(s_i, s_{-i}) \geq v_i(s_i', s_{-i})$, and thus that s_i cannot be a best response to any $s_{-i} \in S_{-i}$. ■

A companion to this proposition would explore strictly dominant strategies, which are in some loose way the "opposite" of strictly dominated strategies. You should easily be able to convince yourself that if a strategy s_i^D is a strictly dominant strategy then it must be a best response to *anything* i's opponents can do. This immediately implies the next proposition, which is slightly broader than the simple intuition just provided and requires a bit more work to prove formally:

Proposition 4.4 *If in a finite normal-form game s^* is a strict dominant strategy equilibrium, or if it uniquely survives IESDS, then s_i^* is a best response to $s_{-i}^* \forall i \in N$.*

Proof If s^* is a dominant strategy equilibrium then it uniquely survives IESDS, so it is enough to prove the proposition for strategies that uniquely survive IESDS. Suppose s^* uniquely survives IESDS, and choose some $i \in N$. Suppose in negation to the proposition that s_i^* is not a best response to s_{-i}^*. This implies that there

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exists an $s_i' \in S_i \backslash \{s_i^*\}$ (this is the set S_i without the strategy s_i^*) such that $v_i(s_i', s_{-i}^*) > v_i(s_i^*, s_{-i}^*)$. Let $S_i' \subset S_i$ be the set of all such s_i' for which $v_i(s_i', s_{-i}^*) > v_i(s_i^*, s_{-i}^*)$. Because s_i' was eliminated while s_{-i}^* was not (recall that s^* uniquely survives IESDS), there must be some s_i'' such that $v_i(s_i'', s_{-i}^*) > v_i(s_i', s_{-i}^*) > v_i(s_i^*, s_{-i}^*)$, implying that $s_i'' \in S_i'$. Because the game is finite, an induction argument on S_i' then implies that there exists a strategy $s_i''' \in S_i'$ that must survive IESDS. But this is a contradiction to s^* being the unique survivor of IESDS. ■

This is a "proof by contradiction," and the intuition behind the proof may be a bit easier than the formal write-up. The logic goes as follows: If it is true that s_i^* was not a best response to s_{-i}^* then there was some other strategy s_i' that was a best response to s_{-i}^* that was eliminated at some previous round. But then it must be that there was a third strategy that was better than both of i's aforementioned strategies against s_{-i}^* in order to knock s_i' out in some earlier round. But then how could s_i^* knock out this third strategy against s_{-i}^*, which survived? This can't be true, which means that the initial negative premise in the proof, that s_i^* was not a best response to s_{-i}^*, must be false, hence the contradiction.

With the concept of a best response in hand, we need to think more seriously about the following question: to what profile of strategies should a player be playing a best response? Put differently, if my best response depends on what the other players are doing, then how should I choose between all the best responses I can possibly have? This is particularly pertinent because we are discussing static games, in which players choose their actions without knowing what their opponents are choosing. To tackle this important question, we need to give players the ability to form *conjectures* about what others are doing. We have alluded to the next step in the claim made earlier, which stated that "a rational player who believes that his opponents are playing some $s_{-i} \in S_{-i}$ will always choose a best response to s_{-i}." Thus we have to be mindful of what a player *believes* in order to draw conclusions about whether or not the player is choosing a best response.

4.3.2 Beliefs and Best-Response Correspondences

Suppose that s_i' is a best response for player i to his opponents playing s_{-i}', and assume for the moment that it is not a best response for any other profile of actions that i's opponents can choose. When would a rational player i choose to play s_i'? The answer follows directly from rationality: he will play s_i' *only when his beliefs about other players' behavior* justify the use of s_i', or in other words when he believes that his opponents will play s_{-i}'.

Introducing the concept of beliefs, and actions that best respond to beliefs, is central to the analysis of strategic behavior. If a player is fortunate enough to be playing in a game in which he has a strictly dominant strategy then his beliefs about the behavior of others play no role. The player's strictly dominant strategy is his best response *independent of* his opponents' play, and hence it is always a best response. But when no strictly dominant strategy exists, a player must ask himself, "What do I think my opponents will do?" The answer to this question should guide his own behavior.

To make this kind of reasoning precise we need to define what we mean by a player's belief. In a well-defined game, the only thing a player should be thinking

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about is what he believes his opponents are doing. Therefore we offer the following definition:

Definition 4.6 A **belief** of player i is a possible profile of his opponents' strategies, $s_{-i} \in S_{-i}$.

Given that a player has a particular belief about his opponents' strategies, he will be able to formulate a best response to that belief. The best response of a player to a certain strategy of his opponents may be unique, as in many of the games we have seen up to now.

For example, consider the Battle of the Sexes. When Chris believes that Alex is going to the opera, his unique best response is to go to the opera. Similarly, if he believes that Alex will go to the football game, then he should go to the game. For every unique belief there is a best response. Similarly recall that in the Cournot game the best choice of player 1 given any choice of player 2 solved the first-order condition of player 1's maximization problem, resulting in the function

$$q_1(q_2) = \frac{90 - q_2}{2}, \tag{4.3}$$

which assigns a unique value of q_1 to any value of q_2 for $q_2 \in [0, 90]$. Hence the function in (4.3) is the best-response function of firm 1 in the Cournot game.

We can therefore think of a rational player as having a *recipe book* that is a list of instructions as follows: "If I think my opponents are doing s_{-i}, then I should do s_i; if I think they're doing s'_{-i}, then I should do s'_i;. . . . " This list should go on until it exhausts all the possible strategies that player i's opponents can choose. If we think of this list of best responses as a plan, then this plan *maps beliefs into a choice of action,* and this choice of action must be a best response to the beliefs. We can think of this as player i's *best-response function*.

There may, however, be games in which for some beliefs a player will have more than one best-response strategy. Consider, for example, the following simple game:

		Player 2		
		L	C	R
	U	3, 3	5, 1	6, 2
Player 1	M	4, 1	8, 4	3, 6
	D	4, 0	9, 6	6, 8

If player 1 believes that player 2 is playing the column R then both U and D are each a best response. Similarly if player 1 believes that player 2 is playing the column L then both M and D are each a best response.

The fact that a player may have more than one best response implies that we can't think of the best-response mapping from opponents' strategies S_{-i} to an action by player i as a *function,* because by definition a function would select only *one action* as a best response (see Section 19.2 of the mathematical appendix). Thus we offer the following definition:

Definition 4.7 The **best-response correspondence** of player i selects for each $s_{-i} \in S_{-i}$ a subset $BR_i(s_{-i}) \subset S_i$ where each strategy $s_i \in BR_i(s_{-i})$ is a best response to s_{-i}.

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That is, given a belief player i has about his opponents, s_{-i}, the set of all his possible strategies that are a best response to s_{-i} is denoted by $BR_i(s_{-i})$. If he has a unique best response to s_{-i} then $BR_i(s_{-i})$ will contain only one strategy from S_i.

4.3.3 Rationalizability

Equipped with the idea of beliefs and a player's best responses to his beliefs, the next natural step is to allow the players to reason about which beliefs to have about their opponents. This reasoning must take into account the rationality of all players, common knowledge of rationality, and the fact that all the players are trying to guess the behavior of their opponents.

To a large extent we employed similar reasoning when we introduced the solution concept of IESDS: instead of asking what your opponents might be doing, you asked "What would a rational player *not do*?" Then, assuming that all players follow this process by common knowledge of rationality, we were able to make some prediction about which strategies cannot be eliminated.

In what follows we introduce another way of reasoning that rules out irrational behavior with a similar iterated process that is, in many ways, the mirror image of IESDS. This next solution concept also builds on the assumption of common knowledge of rationality. However, instead of asking "What would a rational player *not do*?" our next concept asks "What *might* a rational player *do*?" A rational player will select only strategies that are a *best response to some profile of his opponents*. Thus we have

Definition 4.8 A strategy $s_i \in S_i$ is *never a best response* if there are no beliefs $s_{-i} \in S_{-i}$ for player i for which $s_i \in BR_i(s_{-i})$.

The next step, as in IESDS, is to use the common knowledge of rationality to build an iterative process that takes this reasoning to the limit. After employing this reasoning one time, we can eliminate all the strategies that are *never a best response,* resulting in a possibly "smaller" reduced game that includes only strategies that can be a best response in the original game. Then we can employ this reasoning again and again, in a similar way that we did for IESDS, in order to eliminate outcomes that should not be played by players who share a common knowledge of rationality.

The solution concept of rationalizability is defined precisely by iterating this thought process. The set of strategy profiles that survive this process is called the set of *rationalizable strategies*. (We postpone offering a definition of rationalizable strategies because the introduction of mixed strategies, in which players can play stochastic strategies, is essential for the complete definition.)

4.3.4 The Cournot Duopoly Revisited

Consider the Cournot duopoly example used to demonstrate IESDS in Section 4.2.2, with demand $p(q) = 100 - q$ and costs $c_i(q_i) = 10q_i$ for both firms. As we showed earlier, firm 1 maximizes its profits $v_1(q_1, q_2) = (100 - q_1 - q_2)q_1 - 10q_1$ by setting the first-order condition $90 - 2q_1 - q_2 = 0$. Now that we have introduced the idea of a best response, it should be clear that this firm's best response is *immediately derived* from the first-order condition. In other words, if firm 1 believes that firm 2 will choose

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the quantity q_2, then it should choose q_1 according to the best-response function,

$$BR_1(q_2) = \begin{cases} \frac{90-q_2}{2} & \text{if } 0 \leq q_2 < 90 \\ 0 & \text{if } q_2 \geq 90. \end{cases}$$

Notice that the best response is indeed a function. For all $0 \leq q_2 < 90$ there is a unique positive best response. For $q_2 \geq 90$ the price is guaranteed to be below 10, in which case any quantity firm 1 will choose will yield a negative profit (its costs per unit are 10), and hence the best response is to produce nothing. Similarly we can define the best-response correspondence of firm 2, which is symmetric.

Examining $BR_1(q_2)$ implies that firm 1 will choose to produce *only* quantities between 0 and 45. That is, there will be no beliefs about q_2 for which quantities above 45 are a best response. By symmetry the same is true for firm 2. Thus a first round of rationalizability implies that the only quantities that can be best-response quantities for both firms must lie in the interval [0, 45]. The next round of rationalizability for the game in which $S_i = [0, 45]$ for both firms shows that the best response of firm i is to choose any quantity $q_i \in [22.5, 45]$. Just as with IESDS, this process will continue on and on. The set of rationalizable strategies converges to a single quantity choice of $q_i = 30$ for both firms.

4.3.5 The "p-Beauty Contest"

Consider a game with n players, so that $N = \{1, \ldots, n\}$. Each player has to choose an integer between 0 and 20, so that $S_i = \{0, 1, 2, \ldots, 19, 20\}$. The winners are the players who choose an integer number that is closest to $\frac{3}{4}$ of the average. For example, if $n = 3$, and if the three players choose $s_1 = 1$, $s_2 = 5$, and $s_3 = 18$, then the average is $(1 + 5 + 18) \div 3 = 8$, and $\frac{3}{4}$ of the average is 6, so the winner is player 2. This game is called the ***p*-beauty contest** because, unlike in a standard beauty contest, you are not trying to guess what everyone else is guessing (beauty is what people choose it to be), but rather you are trying to guess p times the average, in this case $p = \frac{3}{4}$.[4]

Note that it is possible for more than one person to be a winner. If, for example, $s_1 = s_2 = 2$ and $s_3 = 8$, then the average is 3 and $\frac{3}{4}$ of the average is $2\frac{1}{4}$, so that both player 1 and player 2 are winners. Because more than one person can win, define the set of winners $W \subset N$ as those players who are closest to $\frac{3}{4}$ of the average, and the rest are all losers. The set of winners is defined as[5]

$$W = \left\{ i \in N : \arg\min_{i \in N} \left| s_i - \frac{3}{4}\frac{1}{n}\sum_{j=1}^{n} s_j \right| \right\}.$$

4. John Maynard Keynes (1936) described the playing of the stock market as analogous to entering a newspaper beauty-judging contest in which one selects the six prettiest faces out of a hundred photographs, with the prize going to the person whose selections are closest to those of all the other players. Keynes's depiction of a beauty contest is a situation in which you want to guess what others are guessing.

5. The notation $\arg\min_{i \in N} \left| s_i - \frac{3}{4}\frac{1}{n}\sum_{i=1}^{n} s_i \right|$ means that we are selecting all the i's for which the expression $\left| s_i - \frac{3}{4}\frac{1}{n}\sum_{j=1}^{n} s_j \right|$ is minimized. Since this expression is the absolute value of the difference between s_i and $\frac{3}{4}\frac{1}{n}\sum_{j=1}^{n} s_j$, this selection will result in the player (or players) who are closest to $\frac{3}{4}$ of the average.

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To introduce payoffs, each player pays 1 to play the game, and winners split the pot equally among themselves. This implies that if there are $m \geq 1$ winners, each gets a payoff of $\frac{N-m}{m}$ (his share of the pot net of his own contribution) while losers get -1 (they lose their contribution).

Turning to rationalizable strategies, we must begin by finding strategies that are never a best response. This is not an easy task, but some simple insights can get us moving in the right direction. In particular the objective is to guess a number closest to $\frac{3}{4}$ of the average, which means that a player would want to guess a number that is generally smaller than the highest numbers that other players may be guessing.

This logic suggests that if there are strategies that are never a best response, they should be the higher numbers, and it is natural to start with 20: can choosing $s_i = 20$ be a best response? If you believe that the average is below 20, then 20 cannot be a best response—a lower number will be the best response. If the average were 20, that means that you and everyone else would be choosing 20, and you would then split the pot with all the other players. If you believe this, and instead of 20 you choose 19, then you will win the whole pot for sure, regardless of the number of players, because for any number of players n if everyone else is choosing 20 then 19 will be a unique winner.[6] This shows that 20 can never be a best response. Interestingly 19 is not the *unique* best response to the belief that all others are playing 20. (This is left as an exercise.) The important point, however, is that 20 *cannot* be a best response to *any* beliefs a player can have.

This analysis shows that only the numbers $S_i^1 = \{0, 1, \ldots, 19\}$ survive the first round of rationalizable behavior. Similarly after each additional round we will "lose" the highest number until we go through 19 rounds and are left with $S_i^{19} = \{0, 1\}$, meaning that after 19 rounds of dropping strategies that cannot be a best response, we are left with two strategies that survive: 1 and 0. If $n > 2$, we cannot reduce this set further: if, for example, player i believes that all the other players are choosing 1 then choosing 1 is a best response for him. (This is left as an exercise.) Similarly regardless of n, if he believes that everyone is choosing 0 then choosing 0 is his best response. Thus we are able to predict using rationalizability that players will not choose a number greater than 1, and if there are only two players then we will predict that both will choose 0.

Will this indeed predict behavior? Only if our *assumptions* about behavior are correct. If you were to play this game and you don't think that your opponents are doing these steps in their minds, then you may want to choose a number higher than 1. An interesting set of experiments is summarized in Camerer (2003).[7]

Remark By now you must have concluded that IESDS and rationalizability are two sides of the same coin, and you might even think that they are one and the same. This is almost true, and for two-player games it turns out that these two processes indeed result in the same outcomes. We discuss this issue briefly in Section 6.3, after we introduce the concept of mixed strategies. A more complete treatment can be found in Chapter 2 of Fudenberg and Tirole (1991).

6. The average, $\frac{1}{n}[(n-1)20 + 19]$, lies between 19.5 (if $n = 2$) and 20 (for $n \to \infty$), so that $\frac{3}{4}$ of the average lies between $14\frac{5}{8}$ and 15.

7. From my own experience running these games in classes of students, it is rare that the winning number is below 4.

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4.3.6 Evaluating Rationalizability

In terms of existence, uniqueness, and implications for Pareto optimality, rationalizability is practically the same as IESDS. It will sometimes have bite, and may even restrict behavior quite dramatically as in the examples given. But if applied to the Battle of the Sexes, rationalizability will say "anything can happen."

4.4 Summary

- Rational players will never play a dominated strategy and will always play a dominant strategy when it exists.
- When players share common knowledge of rationality, the only strategies that are sensible are those that survive IESDS.
- Rational players will always play a best response to their beliefs. Hence any strategy for which there are no beliefs that justify its choice will never be chosen.
- Outcomes that survive IESDS, rationalizability, or strict dominance need not be Pareto optimal, implying that players may not be able to achieve desirable outcomes if they are left to their own devices.

4.5 Exercises

4.1 **Prove Proposition 4.1:** If the game $\Gamma = \langle N, \{S_i\}_{i=1}^n, \{v_i\}_{i=1}^n \rangle$ has a strictly dominant strategy equilibrium s^D, then s^D is the unique dominant strategy equilibrium.

4.2 **Weak Dominance:** We call the strategy profile $s^W \in S$ a **weakly dominant strategy equilibrium** if $s_i^W \in S_i$ is a weakly dominant strategy for all $i \in N$, that is, if $v_i(s_i, s_{-i}) \geq v_i(s_i', s_{-i})$ for all $s_i' \in S_i$ and for all $s_{-i} \in S_{-i}$.

a. Provide an example of a game in which there is no weakly dominant strategy equilibrium.
b. Provide an example of a game in which there is more than one weakly dominant strategy equilibrium.

4.3 **Discrete First-Price Auction:** An item is up for auction. Player 1 values the item at 3 while player 2 values the item at 5. Each player can bid either 0, 1, or 2. If player i bids more than player j then i wins the good and pays his bid, while the loser does not pay. If both players bid the same amount then a coin is tossed to determine who the winner is, and the winner gets the good and pays his bid while the loser pays nothing.

a. Write down the game in matrix form.
b. Does any player have a strictly dominated strategy?
c. Which strategies survive IESDS?

4.4 **eBay's Recommendation:** It is hard to imagine that anyone is not familiar with eBay, the most popular auction web site by far. In a typical eBay auction a good is placed for sale, and each bidder places a "proxy bid," which eBay keeps in memory. If you enter a proxy bid that is lower than the current highest bid, then your bid is ignored. If, however, it is higher, then the current bid

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increases up to one increment (say, \$0.01) above the *second highest* proxy bid. For example, imagine that three people have placed bids on a used laptop of \$55, \$98, and \$112. The current price will be at \$98.01, and if the auction ended the player who bid \$112 would win at a price of \$98.01. If you were to place a bid of \$103.45 then the player who bid \$112 would still win, but at a price of \$103.46, while if your bid was \$123.12 then you would win at a price of \$112.01.

Now consider eBay's historical recommendation that you think hard about the value you impute to the good and that you enter your true value as your bid—no more, no less. Assume that the value of the good for each potential bidder is independent of how much other bidders value it.

a. Argue that bidding more than your valuation is weakly dominated by actually bidding your valuation.
b. Argue that bidding less than your valuation is weakly dominated by actually bidding your valuation.
c. Use your analysis to make sense of eBay's recommendation. Would you follow it?

4.5 **Iterated Elimination:** In the following normal-form game, which strategy profiles survive iterated elimination of strictly dominated strategies?

		Player 2		
		L	C	R
	U	6, 8	2, 6	8, 2
Player 1	M	8, 2	4, 4	9, 5
	D	8, 10	4, 6	6, 7

4.6 **Roommates:** Two roommates each need to choose to clean their apartment, and each can choose an amount of time $t_i \geq 0$ to clean. If their choices are t_i and t_j, then player i's payoff is given by $(10 - t_j)t_i - t_i^2$. (This payoff function implies that the more one roommate cleans, the less valuable is cleaning for the other roommate.)

a. What is the best response correspondence of each player i?
b. Which choices survive one round of IESDS?
c. Which choices survive IESDS?

4.7 **Campaigning:** Two candidates, 1 and 2, are running for office. Each has one of three choices in running his campaign: focus on the positive aspects of one's own platform (call this a positive campaign [P]), focus on the positive aspects of one's own platform while attacking one's opponent's campaign (call this a balanced campaign [B]), or finally focus only on attacking one's opponent (call this a negative campaign [N]). All a candidate cares about is the probability of winning, so assume that if a candidate expects to win with probability $\pi \in [0, 1]$, then his payoff is π. The probability that a candidate wins depends on his choice of campaign and his opponent's choice. The probabilities of winning are given as follows:

If both choose the same campaign then each wins with probability 0.5.

If candidate i uses a positive campaign while $j \neq i$ uses a balanced one then i loses for sure.

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If candidate i uses a positive campaign while $j \neq i$ uses a negative one then i wins with probability 0.3.

If candidate i uses a negative campaign while $j \neq i$ uses a balanced one then i wins with probability 0.6.

a. Model this story as a normal-form game. (It suffices to be specific about the payoff function of one player and to explain how the other player's payoff function is different and why.)
b. Write down the game in matrix form.
c. What happens at each stage of elimination of strictly dominated strategies? Will this procedure lead to a clear prediction?

4.8 **Beauty Contest Best Responses:** Consider the p-beauty contest presented in Section 4.3.5.

a. Show that if player i believes that everyone else is choosing 20 then 19 is not the only best response for any number of players n.
b. Show that the set of best-response strategies to everyone else choosing the number 20 depends on the number of players n.

4.9 **Beauty Contest Rationalizability:** Consider the p-beauty contest presented in Section 4.3.5. Show that if the number of players $n > 2$ then the choices $\{0, 1\}$ for each player are both rationalizable, while if $n = 2$ then only the choice of $\{0\}$ by each player is rationalizable.

4.10 **Popsicle Stands:** Five lifeguard towers are lined up along a beach; the leftmost tower is number 1 and the rightmost tower is number 5. Two vendors, players 1 and 2, each have a popsicle stand that can be located next to one of five towers. There are 25 people located next to each tower, and each person will purchase a popsicle from the stand that is closest to him or her. That is, if player 1 locates his stand at tower 2 and player 2 at tower 3, then 50 people (at towers 1 and 2) will purchase from player 1, while 75 (from towers 3, 4, and 5) will purchase from vendor 2. Each purchase yields a profit of \$1.

a. Specify the strategy set for each player. Are there any strictly dominated strategies?
b. Find the set of strategies that survive rationalizability.

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